PlantCare+

Specification			
Version	1	Date:	15/04/2026
Revision			
1	First Issue	15/04/2026	

PART A

User Manual

Group 15

1. Introduction

This document acts as the user manual and technical datasheet for the Plant Care+ System, a user focused self-contained embedded system designed to maintain plant health with minimal user support. The system is designed to continuously monitors its environment and take necessary precaution and corrective steps to support plant health in the following ways:

- Automatic soil watering
  - Reads soil moisture and activates the embedded watering system depending on specific
- Water tank monitoring
  - Continuously checks the tank level ensuring pump levels are satisfactory
- Plant tilt correction via servo rotation
  - Pressure sensors hidden at the base detect plant and rotates the platform to restore an upright orientation and ensure straight growth.

This guide is designed to support users in the setup, Day to Day and troubleshooting process.

2. General Specifications

Parameter	Value / Detail
Microcontroller	ESP32 (Xtensa dual-core 32-bit LX6 processor, up to 240 MHz, integrated Wi-Fi 802.11 b/g/n and Bluetooth v4.2 + BLE)
Power supply	3.3 V via USB
Soil moisture sensor	Capacitive probe; 0–100% mapped output
Water level sensor	Water level sensor mounted in tank
Water pump	Submersible DC pump
Rotating base	Servo motor (standard PWM control, 180° range)
Tilt detection	Pressure sensors under base platform
Operating temperature	5 °C – 40 °C (indoor use)

3. Setup and Getting Started

3.1 Component Checklist

Before beginning setup, confirm that all the following components are present:

Parameter	Value / Detail
Main control unit	ESP32 development board with system design preloaded
Soil moisture sensor	Probe with wires; 1× per plant
Water level sensor	Mounted inside water tank
Water pump + tubing	Submersible pump with clear tubing that can be cut to users desired length
Rotating base	Platform with servo motor and ball bearings integrated
Pressure sensors	4 x Ohm-Mite 14-21 pressure gauges (pre-fitted)
Water tank	Container 1L capacity
Base Power supply	4x AAA battery pack
Tank power supply	2x AAA battery pack

3.2 Physical Assembly

Assemble the system in the following order.

1. Place the base on a flat, stable surface. Ensure all pressure sensors make even contact with the ground. Secure the water level sensor in the designated mounting point.
2. Then place the rotating base, water tank, pot exterior and pot interior on top.
3. Set the plant pot firmly into the pot interior.
4. Insert the soil moisture sensor into the soil - approximately halfway into the pot. The probe must not touch the pot walls or base.
5. Fill the water tank.
6. Route the pump tubing from the water tank to the soil surface then feed the outlet so it waters into the soil.
7. Verify all wiring connections between sensors, pump, servo motor, and the ESP32 are secure,
8. Connect the power supply and turn on.

NOTE	Keep all electronic components and connectors away from direct contact with water or soil. Only the sensor probes are designed for wet contact.

3.3 Initial Run Sequence

On first power-on, the system runs an initialisation routine.

Then the soil moisture will take initial baseline readings, the water level confirms the tank level status, The pressure sensors then record the baseline weight distribution for the plants current position. The system has an AUTO mode which begins the continuous monitoring process.

NOTE	Do not reposition the plant as the plants position is used for tilt referencing and all subsequent rotating decisions. Once the plant is moved, the pressure sensors baseline needs to be reset.

4. System Operation

Once initialised, the system runs a continuous closed-loop cycle. Each iteration performs the following steps in sequence:

4.1 Soil Moisture Monitoring

The ESP32 is designed to read the analogue output of the soil moisture sensor and convert the raw ADC value into a moisture percentage. These values are compared against calibrated thresholds, 0% is dry and 100% is wet.

Parameter	Value / Detail
Watering threshold	< 12% moisture
Sensor read interval	Every 1 second
Output	Turns the pump on when moisture is below threshold, provided the tank has sufficient water and time elapsed since last watering has passed.

NOTE	The sensor is designed to monitor soil against the principle of capacitive detection, so different soil moistures, compactness of soil and insertion effect this. Recalibration is required if soil mixture changes.

4.2 Water Tank Level Check

The pump is designed to check tank levels before activating to prevent damaging the pump and creating inaccurate soil readings in the future,

- tank levels are read below the allowance; the pump is disabled.

4.3 Automatic Watering

When both conditions are met, the pump is activated for a fixed duration.

Parameter	Value / Detail
Activation condition	Moisture < threshold AND tank level OK
Pump run duration	5 seconds per cycle
lockout period	10 seconds after watering

The system continuously checks moisture in a loop; it cannot start another watering cycle until the lockout period has expired.

4.4 Tilt Detection

Four pressure sensors distributed under the rotating base measure the weight difference and weight distribution of the plant. The ESP32 reads all sensor values and an angle to rotate. If the weight is shifted beyond the imbalanced threshold, then the ESP compute an angle to rotate in the opposite direction.

Parameter	Value / Detail
Detection method	Multi-point pressure sensing, centre of mass calculated,
Imbalance load (kPa)	Set to 2 – minimum total pressure needed for calculation
Output	Angle rotation

4.5 Servo Motor

The servo receives the desired rotation amount, the base responds and rotates to the corresponding position.

Parameter	Value / Detail
Rotation Method	Direct positioning to the target angle
Range	5 to 175 degrees
Updated Interval	Every 100 ms
Post-watering behaviour	Baseline is recalibrated after each watering event

Note - System is designed to recalibrate after watering as watering disturbs the weight distributions.

4.6. Safety Notes

Parameter	Value / Detail
Pump dry run	Ensure tank is full. Pump is disabled when water levels are low and user must refill as to not damage the pump.
Electronics and water	All components besides pump, soil moisture and water level sensor must be kept clear of moisture.
Rotating base load	Do not exceed the weight rating of the servo motor. Excessive load will stall or damage the servo.
Power supply	Use the serial cable provided to attach to personal device.

4.7. Known Limitations

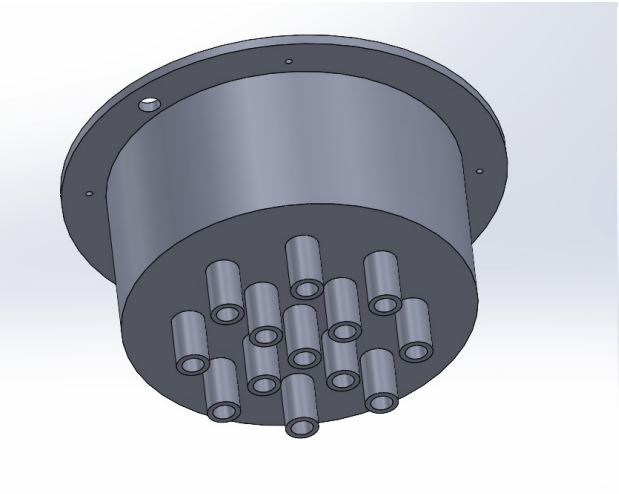
Parameter	Value / Detail
Soil type variation	Soil composition can heavily skew soil readings, recalibration is recommended when changing soil.
Watering	The tube is designed to deliver water to one point therefore the design is susceptible to uneven saturation. Manual watering may need to be supplements.
Small tilt angles	Gradual leans may not be detected by the sensors; the sensors are designed to detect large enough pressure differences. The tilt may only be detected in the plant when large enough
Servo limit	The base can only rotate within the servo range (max 180°). Plants requiring correction beyond this range will need manual repositioning. The base motor has a range of 180 degrees and a correction beyond this range would be out of scope.
No remote monitoring	Version 1.0 operates standalone with no app, user interface or monitoring abilities yet available.
Surface dependance	The pressure sensors accuracy is dependent on the surface its placed on. Uneven surfaces reduce the tilt accuracy.

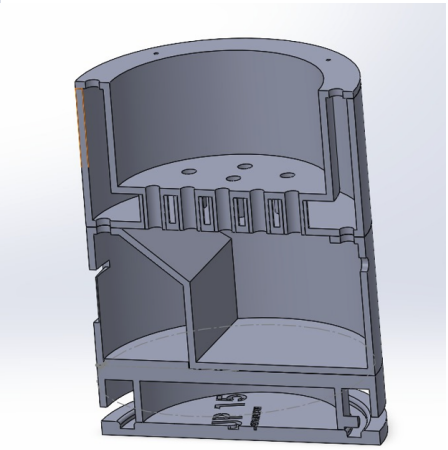
Firmware Parameters

The following can be adjust to suit plant needs:

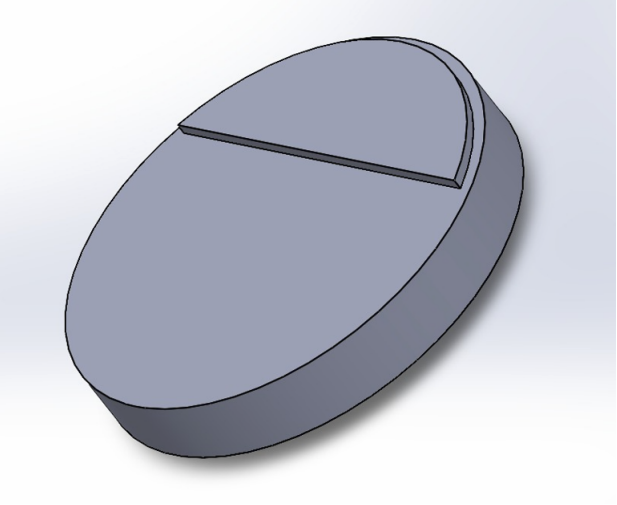
Parameter	Value / Detail
Moisture threshold (%)	Set to 12 - Lower to water less frequently and raise to depending on plant species.
Pump run duration (ms)	Set to 5000 ms. Increase if the pot is large or dry soil type
Lockout period (s)	Set to 10 - can be extended for slow draining soil.
Imbalance load (kPa)	Set to 2.0 – change based on desired tilt sensitivity
Loop update	100 ms loop cycle

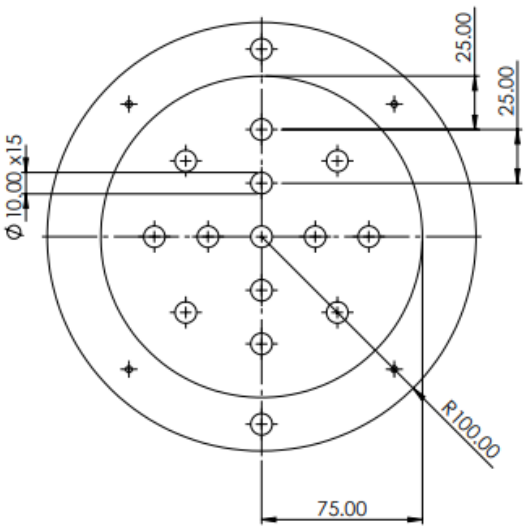
Technical Drawings

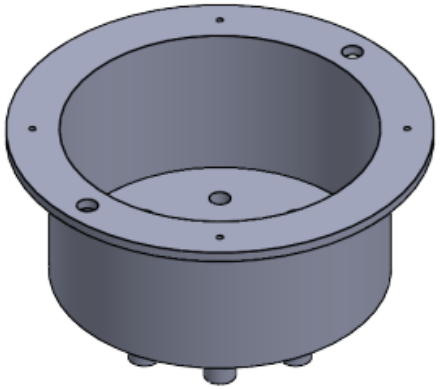


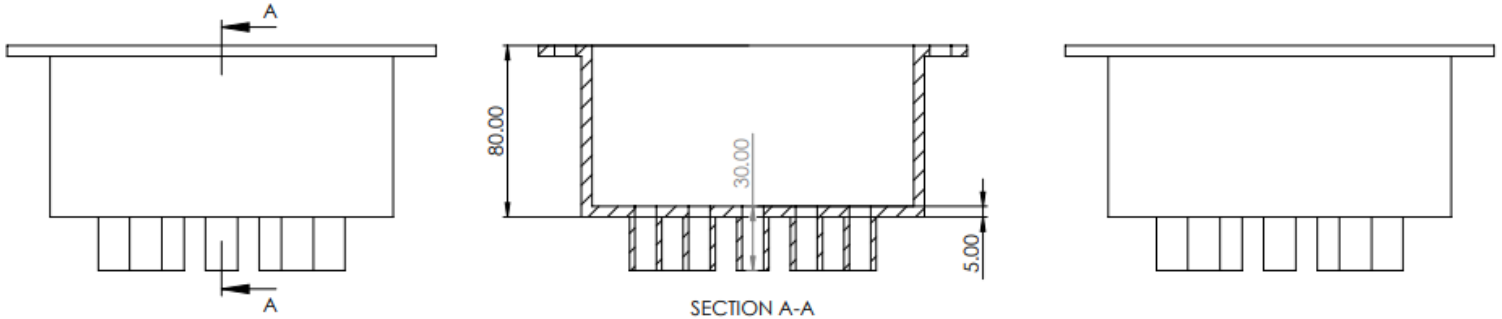








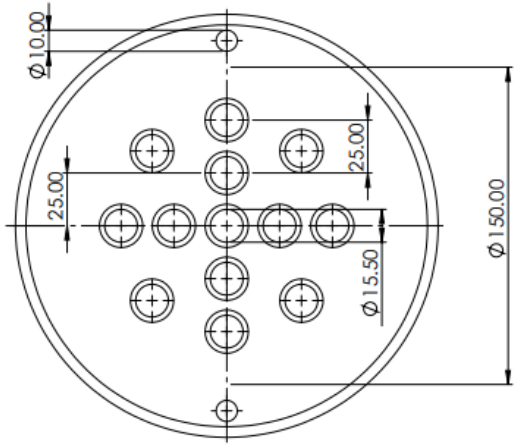


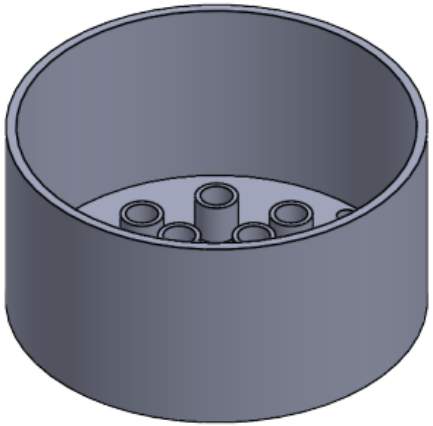


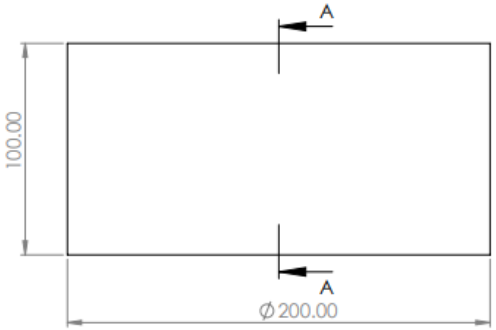
SECTION A-A

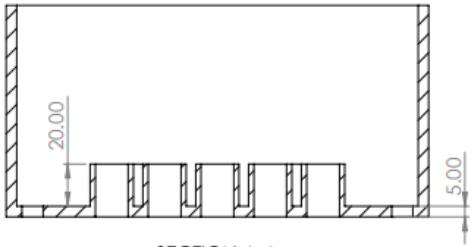
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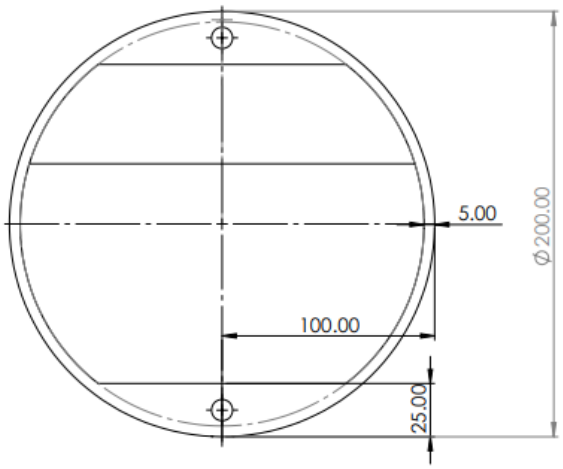


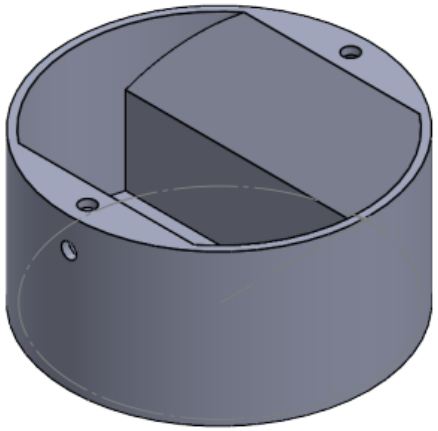
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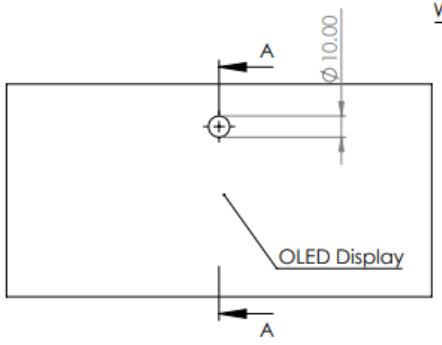


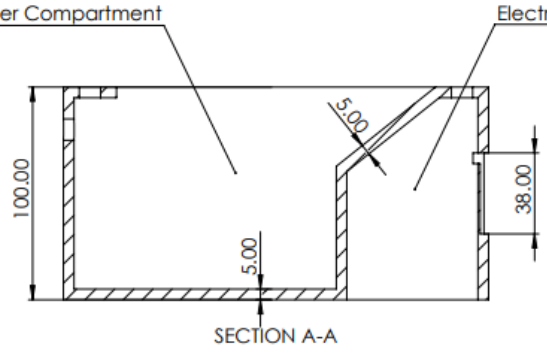
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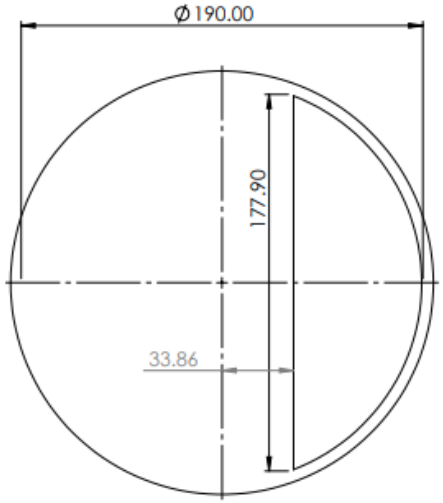


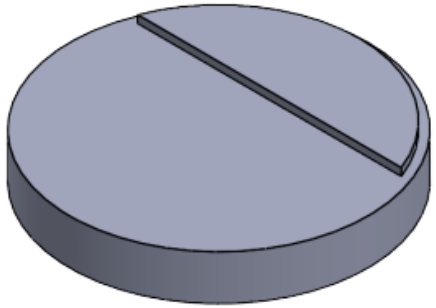


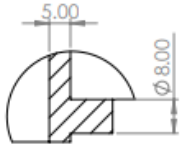


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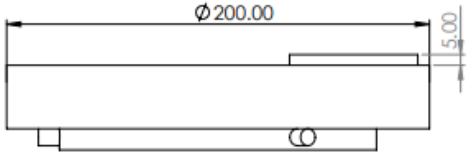
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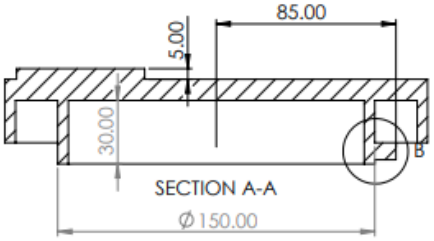




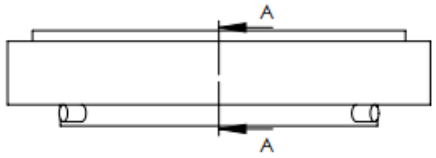


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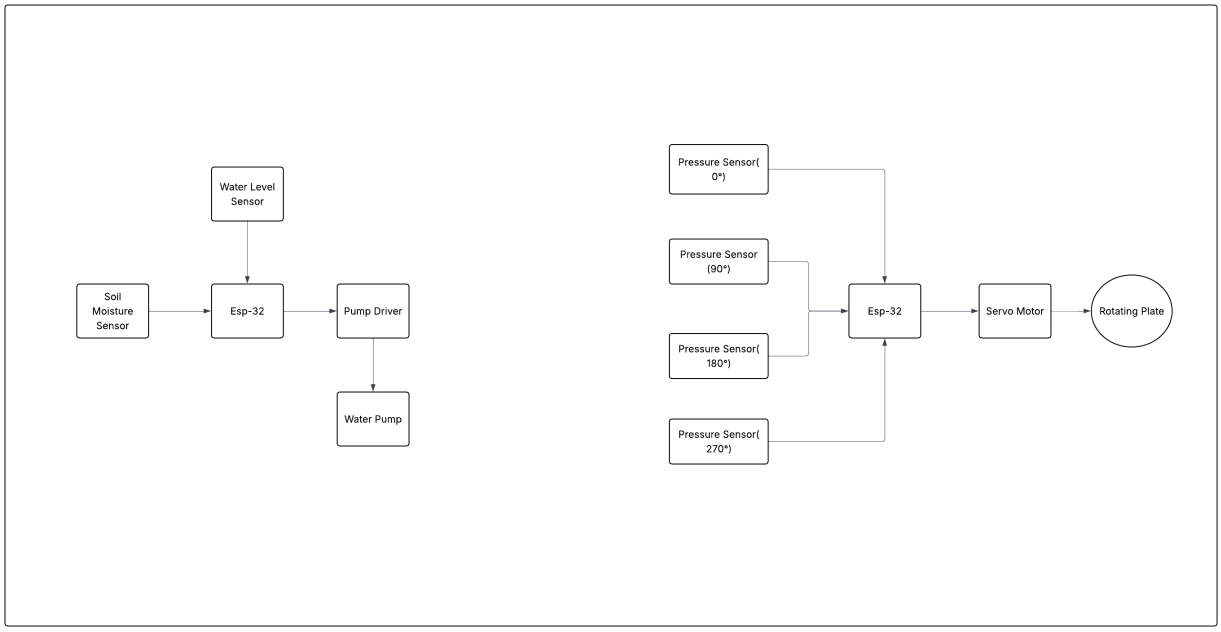
PART B

Technical Data Sheet

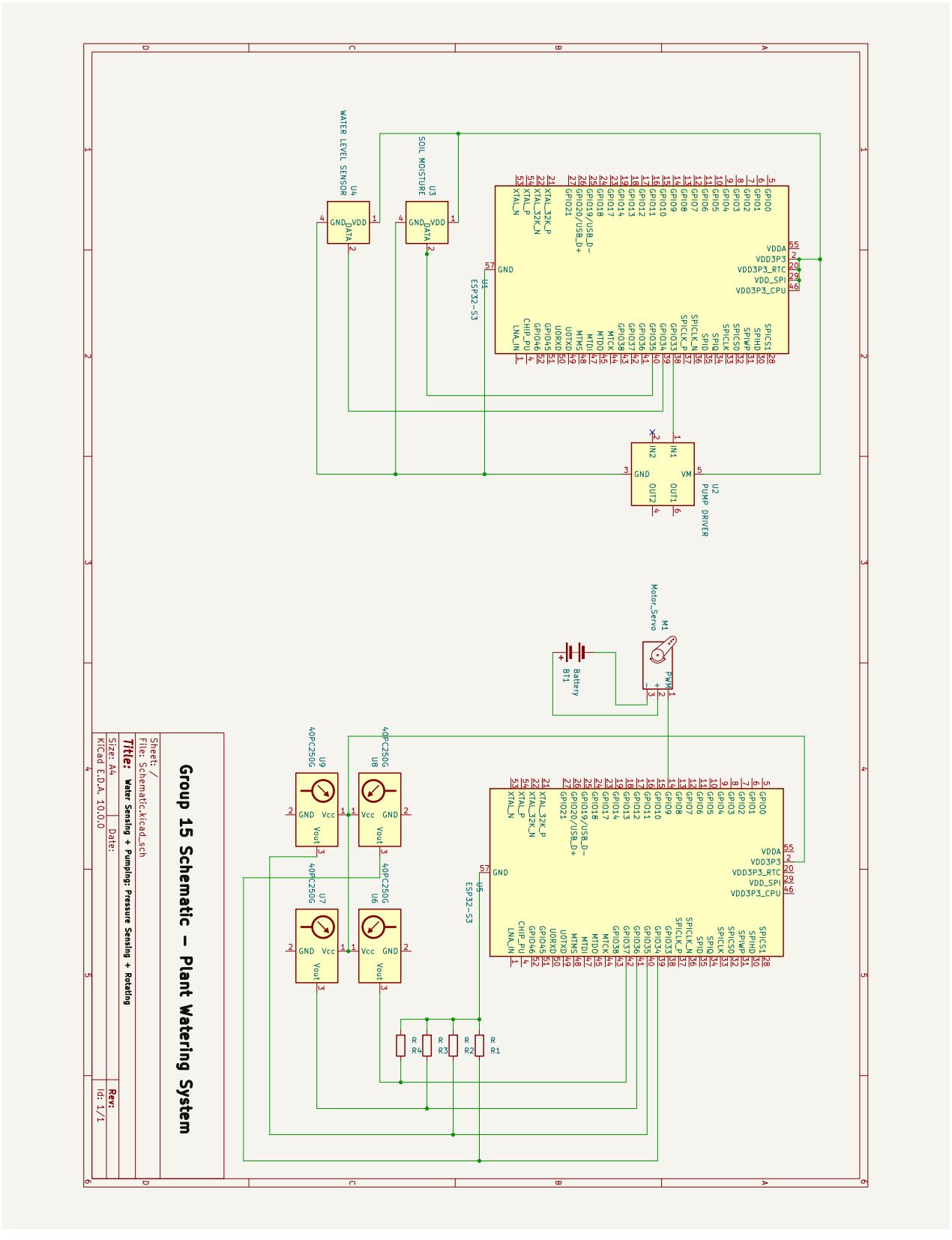
1. System Overview

The PlantCare+ is embedded control system for plant health management. It is designed to automatically maintain plant health by monitoring environmental conditions and responding without user intervention. It operates as a closed loop control system, using two ESP32 microcontrollers to integrate the modular subsystems of the device. These include: soil moisture sensing, water-level monitoring, pressure sensing and actuation. The sensors are monitored in real time, and use continuous feedback as well as threshold logic for the system response. The system uses 3.3V logic supply to the ESP via 2x AAA battery pack and has separate power for pump and servo motor. The system requires minimal user interaction asides from set up and water refill.

2. Block Diagram



3. Circuit Diagram



4. Components

Component	Model	Function	Qty	Key Specs
Esp32 dev bord	ESP32-WROOM-32	Main control, sensor processing, actuator control	2	3.3V logic, ADC, PWM, Wi-Fi
Soil moisture sensor	EN0193 DFRobot	Soil moisture measurement	1	Analog output, corrosion resistant
Water level sensor	SEN0205 DFRobot	Tank level detection	1	Digital/Analog threshold output
Pressure sensors	FSR04BE OHMITE	Load distribution/ plant tilt detection	4	Resistive pressure sensing
Servo motor	MG996R	Rotating base actuation	1	PWM control; 180 degree range of motion
DC water pump		Irrigation actuation	1	DC supply, fixed duration pumping
USB supply		Logic power supply for Esp32	1	5V USB input, regulated 3.3 board logic
9V Battery		External actuator power supply	1	9V nominal output
Driver interface	Adafruit DRV8871	Interface between pump and Esp32	1	6.5-45V, Peak current: 3.6A, Logic control (PWM/digital)

5. Sensor Interface and Calibration

The signals are interfaced to the Esp32 GPIO pins, including both analog (via ADC conversion) and digital inputs. Signals operate in a 0-3.3V range. The signal conditioning and mapping are applied to the system using embedded software (Arduino framework) with periodic sampling to support the real-time control decisions.

5.1 Soil Moisture Sensor

Sensor Type: Capacitive soil moisture sensor

Interface: Analog output connected to ESP32 ADC pin

Signal Range: 0-3.3V corresponding to variation in soil moisture

Signal Processing:

- ADC conversion using ESP32 (12-bit resolution, 0–4095 range)
- ADC values mapped to percentage moisture (0–100%)
- Sensor exhibits inverse response (higher output → lower moisture content)

Calibration:

- Dry soil reading used as minimum reference value
- Saturated soil reading used as maximum reference value
- Linear scaling applied between calibrated limits

Sampling:

- Periodic sampling at approximately 1 second intervals
- Continuous updates integrated into control loop

Control integration:

- Moisture level compared against predefined threshold (30%)
- Primary condition for automatic watering activation
- Combined with water level sensor status before pump actuation

5.2 Water Level Sensors

Sensor Type: Water level sensor (mounted in tank)

Interface: Digital input connected to ESP32 GPIO pin 26

Signal Processing:

- Digital input read as discrete state (tank empty / sufficient level)
- If analog: threshold comparison applied to determine valid water level

Sampling:

- Periodic sampling integrated with control loop
- Checked prior to pump activation

Control integration:

- Acts as a safety condition for pump operation
- Pump disabled if water level is below minimum threshold
- Prevents dry running and potential pump damage

5.3 Pressure Sensors

Sensor type: Resistive pressure sensors (Ohm-Mite 14-21) ×4

Interface: Analog inputs connected to ESP32 ADC pins

Signal range: Variable voltage output via voltage divider configuration

Signal processing:

- Each sensor produces an analog voltage proportional to applied load
- ADC conversion (12-bit resolution, 0–4095 range)
- Sensor values compared to determine weight distribution

Calculation:

- Average sensor value calculated across all four sensors
- Deviation of each sensor from average used to detect imbalance
- Direction and magnitude of tilt inferred from relative differences

Sampling:

- Periodic sampling (synchronised across all sensors)
- Continuous monitoring for real-time tilt detection

Control integration:

- Imbalance threshold applied
- Trigger for rotational correction via servo motor
- System recalibrates baseline after watering events

Limitations:

- Sensitivity to uneven surface placement
- Drift over time requiring recalibration
- Accuracy dependent on consistent load distribution and sensor positioning

6. Control Logic

The control architecture was implemented using ESP32 microcontrollers for continuous loop-based operation with periodic sensor sampling. Due to physical separation and wiring constraints, it was necessary to distribute the architecture between two microcontrollers.

6.1 Sensor Acquisition

ADC resolution: 12-bit (0 - 4095)

Analog inputs:

- soil moisture sensor
- pressure sensors ×4

Digital inputs:

- water level sensor

Sampling method:

- averaged ADC sampling used to reduce noise
- periodic sampling intervals:
  - watering loop: ~1 s
  - rotation loop: ~100 m

6.2 Soil Moisture and Watering Logic

Soil moisture mapped from calibrated ADC range to 0–100%

Hysteresis-based control:

- watering initiated below lower threshold (e.g., 30%)
- watering demand OFF above upper threshold (e.g., 35%)

Activation conditions:

- soil moisture below lower threshold
- water reservoir status valid
- pump not in lockout state

Pump Actuation

- fixed-duration actuation via motor driver
- pulse-based operation
- pump runtime = 3000ms per cycle

Lockout Control

- delay applied after pump cycle before next evaluation to allow moisture to stabilise within soil

Control Behaviour

- repeated pump cycles while moisture remains below upper threshold
- watering terminated once moisture exceeds upper threshold

6.3 Tank Monitoring

Water level sensor is interfaced as digital input to ESP32 GPIO pin. The signal is interpreted as two binary states:

- sufficient water level
- low reservoir

Control Integration

- used as enable condition for the pump operation
- pump activation only permitted when the reservoir status is valid

System Role

- prevents dry running and potential damage to pump
- acts as a safety within the watering control loop

6.4 Pressure-Based Tilt Detection

The four FSR (force sensitive resistors) sensors were positioned at 0°, 90°, 180°, 270° around base.

Interface:

- analog inputs to ESP32 ADC pins
- voltage divider configuration used to convert resistance change to measurable voltage

Signal processing:

- ADC conversion (12-bit, 0–4095 range)
- raw readings converted to estimated pressure values
- moving average filtering applied to reduce noise

V**alidity check:**

- total pressure calculated across all sensors
- minimum threshold applied to ensure valid contact condition

Tilt detection:

- relative pressure distribution used to identify imbalance
- comparison between opposing sensor pairs (0°–180°, 90°–270°)

Deadband implementation

- small deviations below threshold ignored
- prevents false tilt detection due to noise or minor disturbances

System role

- provides input for rotational correction control
- enables detection of plant tilt and orientation imbalance

6.5 Rotational Control Logic

Rotational control implemented using servo actuation driven by ESP32 PWM output.

Input condition

- tilt condition detected from pressure imbalance exceeding threshold

Control strategy:

- threshold-based correction
- baseline referenced imbalance detection

Correction method:

- incremental rotational adjustment applied
- system re-samples pressure sensors after each movement step
- correction continues until imbalance falls within acceptable bounds

Control considerations:

- plant growth and tilt occur over long time scales
- system designed for slow correction
- high-speed or continuous adjustment not required

6.6 Actuator Control

Pump:

- digital switching from ESP32
- fixed-duration pulse-based actuation

Servo:

- PWM-controlled output (~50 Hz)
- angular positioning within constrained range
- updated only under valid tilt condition

7. Testing and Results

7.1 Soil Moisture Sensor Testing

The sensor output was recorded under dry and saturated soil conditions The calibration range established:

- dry soil: ~3500 ADC
- saturated soil: ~1500 ADC

These values were mapped linearly to 0–100% moisture. The sensor response was observed to be stable under static conditions, with only minor variation detected due to soil compaction and probe placement. While not a precise measurement this level of accuracy was sufficient for the irrigation triggering in the control system.

7.2 Watering System Performance

The pump activation was tested under varying soil moisture levels. The trigger condition was verified at ~30% moisture threshold. A fixed-duration actuation of 3000 ms evaluated through the following observations:

- a single pump cycle increased moisture by approximately **5–10%**
- multiple cycles required for significantly dry soil
- lockout delay prevented rapid re-triggering

This pulse-based watering achieved gradual and controlled moisture increase, avoiding overwatering and excessive use of the pump.

7.3 Water Level Sensor Testing

The water level sensor was tested under full and empty reservoir conditions. It was verified that the digital output correctly identified the tank status

- LOW signal corresponded to sufficient water level
- HIGH signal corresponded to low reservoir condition
- Pump correctly disabled when reservoir was empty

These results confirmed the operation of the binary detection was suitable for the system.

7.4 Pressure Sensor and Tilt Detection Testing

The FSR sensors were tested under varying load distributions. The baseline readings were recorded under balanced conditions. There was a measurable variation in the sensor output with applied load. The imbalance detection was consistent across opposite sensor pairs. There was some noise present in the raw readings which was handled through averaging. Once this was implemented the system was able to successfully detect tilt conditions above a defined threshold.

7.5 Rotational Control Testing

The servo actuation was tested under simulated tilt conditions in order to evaluate the performance of the incremental correction behaviour. The system was observed to perform step-based rotation until balance was restored. Stable operation was achieved with no oscillation, effectively restoring the plants alignment.

9. Power Analysis

The system operates using separate power sources:

- 3.3V logic supply for ESP32 and sensors (via USB)
- 9V battery supply for actuators

The separation of logic and load supplies prevents voltage drops and noise affecting control signals

9.1 System Power Consumption

Idle State:

- Dominated by ESP32 and sensors
- Estimated total: 100-150mA

Active watering:

- Pump enabled, additional 200-500 mA used
- Short duration (~3s per cycle)

Rotation event

- Servo actuation results in peak current draw (~500mA)
- This is infrequent due to system design

9.2 Efficiency Considerations

Several design measures in the system were considered to improve the energy efficiency of the system. The low frequency sampling operation (1s loop) reduces the processing load. The pulse-based pump control reduces continuous power consumption. In addition to this, the event-driven actuation with lockout delays minimizes unnecessary energy consumption. Signal averaging also minimizes false triggering of modes due to noise, ensuring actuation only occurs when necessary.

9.3 Limitations

The 9V battery has limited current capability for motor loads. Voltage drop under load may reduce the actuator performance. It wouldn't be suitable for long-term continuous operation without replacement.

PART C

Extensible System

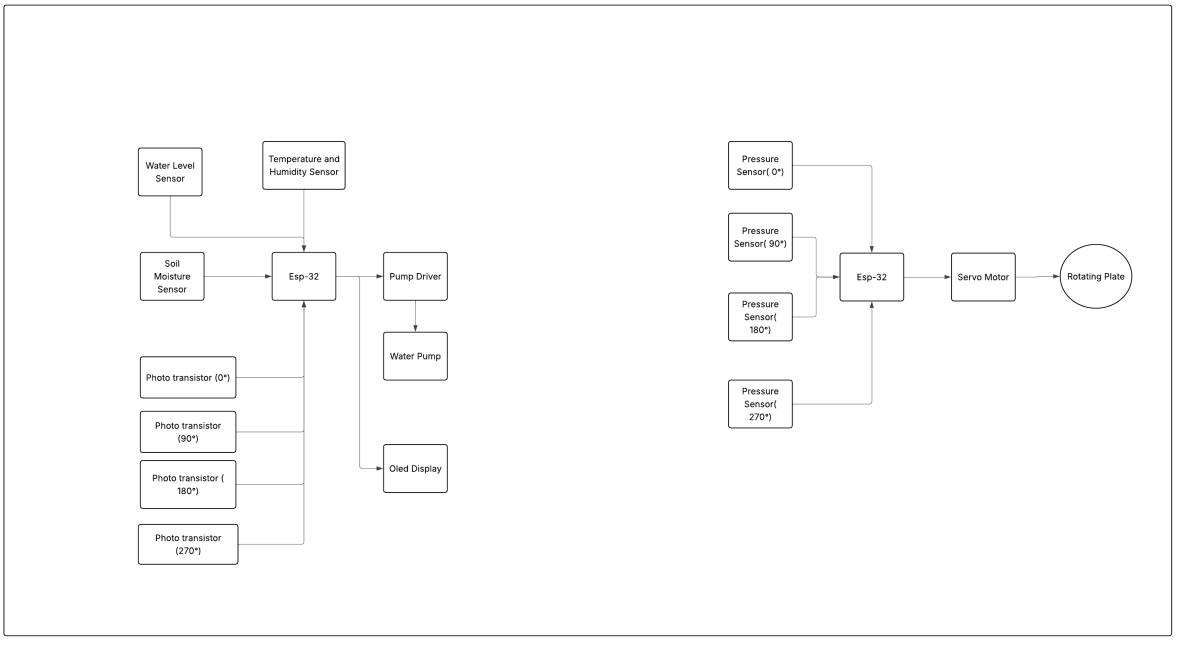
1. Extensible Modules Overview

PlantCare+ is available with several additional modules which can be configured with the standard PlantCare+ device. These modules sit alongside the current design and are not required for operation of the device; however, they add functionality which may benefit certain users.

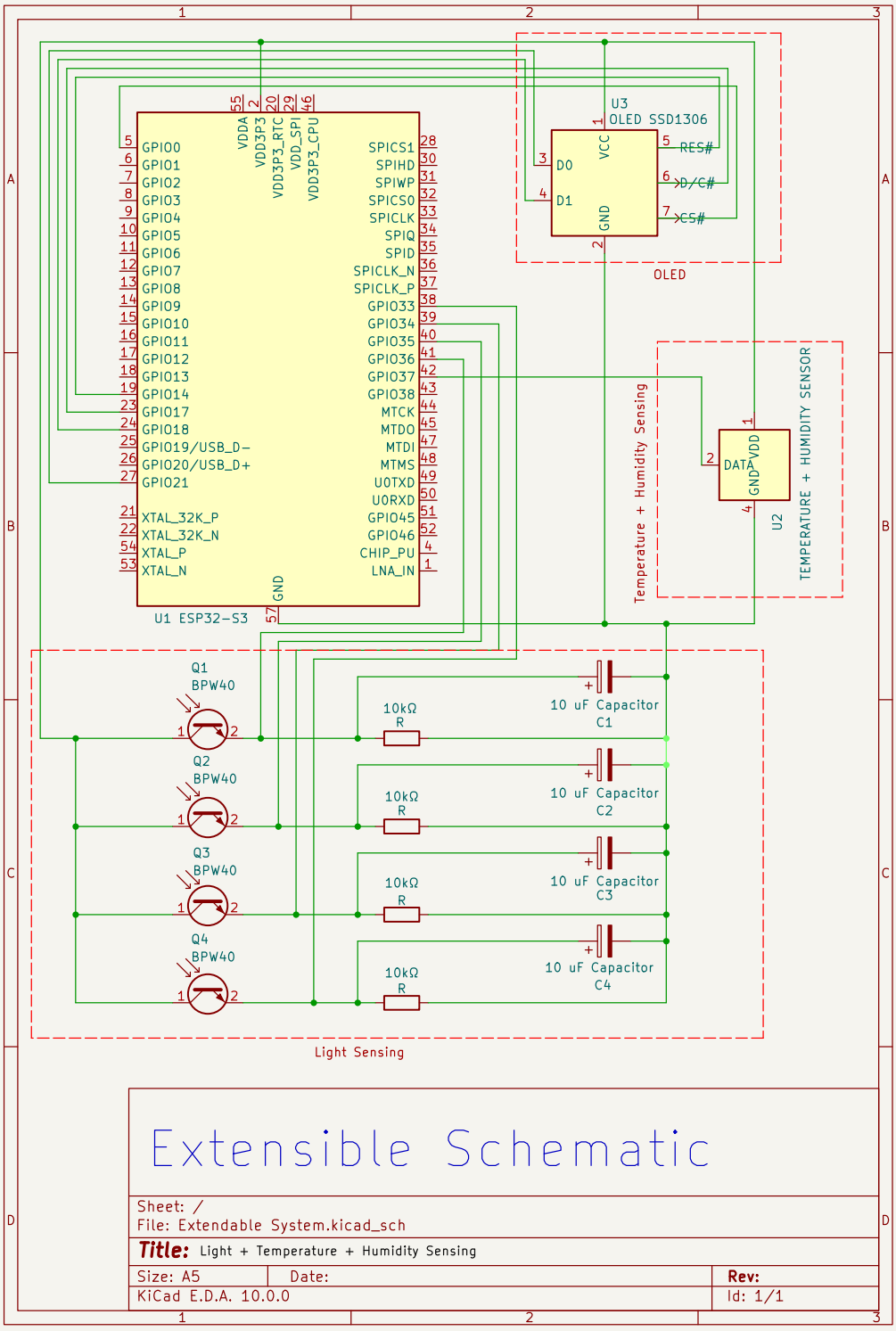
The current extensible modules available for PlantCare+:

1. Light intensity tracking
2. Temperature and Humidity sensing
3. OLED display suite
4. Wi-Fi communication

2. Extensible Block Diagram



3. Extensible Modules Schematic



4. Extensible Components

Module	Attaches to	Component	Model	Function	Qty	Key Specs	Prerequisites Required
Light Intensity Tracking	Top of plant pot	Phototransistor	2-terminal NPN BJT	Vary voltage depending on light intensity	4		Standard PlantCare+ Device; Wi-Fi Compatibility
		Resistor	Standard	Protective + allows capture of voltage division by ESP32	4	10kΩ	
		Capacitor	Standard Polarised	Smooths the output to ESP32	4	10µF	
Temperature + Humidity Sensing	Top of plant pot	Dedicated Temperature + Humidity Sensor	DFRobot DHT11	Outputs ADC value for temperature + humidity	1		Standard PlantCare+ Device; OLED Display Suite
OLED display suite	Front of plant pot	OLED screen	SSD 1306 SPI OLED	Display statistics and information about plant	1		Standard PlantCare+ Device
		Button	Standard for 3.3v	Allows user inputs within menus	2		
		Potentiometer	Standard for 3.3v	Allows user to define precise angle for manual rotation	1		
Wi-Fi Compatibility		Propietary to ESP32 modules used		Allows for communication between ESP32's			Standard PlantCare+ Device; Light Intensity Tracking; OLED display suite

5. Extensible Module Description

5.1 Light Intensity Tracking

The Light Intensity Tracking Module is an optional sensing and orientation-control subassembly designed for use with standard PlantCare+ devices equipped with Wi-Fi communication capability.

Function

The module detects the direction of maximum incident light in the local plant environment and provides this information to the rotating base. The rotating base then adjusts plant orientation in response to measured light conditions and tilt status.

Features

- Compatible with standard PlantCare+ devices with Wi-Fi connectivity.
- Top-mounted configuration for ambient light-direction sensing.
- Uses 4 phototransistors to measure relative incident light intensity from multiple directions.
- Calculates the direction of maximum detected light intensity.
- Communicates light-direction data to the rotating base for orientation control.
- Supports automatic plant alignment toward the dominant light source during normal operation.
- Supports tilt-compensation control by rotating the plant approximately 180∘180∘ opposite to the direction of maximum light intensity when lean is detected.

Operating Principle

The sensor module continuously monitors relative light intensity using four phototransistors. Based on these measurements, the control system estimates the direction corresponding to the highest incident light level. This directional data is transmitted to the rotating base, which positions the plant accordingly.

If the rotating base detects a tilt condition, the control system commands a compensating rotation such that the plant is oriented approximately 180∘180∘ away from the direction of greatest light intensity. This function is intended to reduce asymmetric growth associated with prolonged directional light exposure.

Integration

- Mechanical interface: top-mounted module integrated into the PlantCare+ enclosure.
- Electrical interface: communication with the rotating base through the PlantCare+ control system.
- System requirement: compatible PlantCare+ device with Wi-Fi support.

Use-case

This module is intended to support automated orientation adjustment and improved growth symmetry in environments with non-uniform lighting.

5.2 Temperature + Humidity Sensing

The Temperature + Humidity Sensing module is designed for integration with the standard PlantCare+ device with OLED Display Suite and is mounted at the top of the plant pot. It uses a DFRobot DHT11 sensor to measure ambient temperature and humidity.

Function
Measures ambient temperature and humidity and outputs this data to the PlantCare+ system for environmental monitoring.

Features

- Compatible with the standard PlantCare+ device with OLED Display Suite
- Top-mounted sensor configuration.
- Combined temperature and humidity measurement.

- Uses a DFRobot DHT11 sensor.

Operating Principle
The sensor monitors the surrounding air and sends temperature and humidity data to the control system for processing.

Integration

- Mounting location: top of plant pot.
- Electrical interface: connected to the upper plant pot controller.
- Prerequisite: OLED Display Suite

Use-Case
Intended for more advanced monitoring of plant environmental conditions as part of the PlantCare+ system.

5.3 OLED Display Suite

The OLED Display Suite is a user-interface module designed for integration with the standard PlantCare+ device. It is mounted on the front of the plant pot and uses an SSD1306 SPI OLED, two input buttons, and a potentiometer to display plant data and support user interaction.

Function
Displays plant statistics and general system information, while also providing a local interface for menu navigation and manual control.

Features

- Front-mounted display and control interface.
- Uses an SSD1306 SPI OLED screen.
- Includes two buttons for menu input.
- Includes one potentiometer for precise angle selection during manual rotation.
- Displays plant information such as soil moisture percentage, water tank level, and time since last watered.
- Can display light intensity data and temperature/humidity data when those modules are enabled.

- Supports a manual rotation mode when Wi-Fi functionality is enabled.

Operating Principle
The OLED module receives plant and system data from the PlantCare+ controller and presents it through on-screen menus. User inputs from the buttons and potentiometer are used to navigate menus and adjust settings such as manual rotation angle.

Integration

- Mounting location: front of plant pot.
- Display component: SSD1306 SPI OLED.
- User inputs: two buttons and one potentiometer.
- Prerequisite: standard PlantCare+ device.

Use-Case
Intended to provide the user with a simple on-device interface for viewing plant status, monitoring enabled modules, and accessing manual control functions.

5.4 Wi-Fi Compatibility

The Wi-Fi Compatibility module enables wireless communication between the ESP32 modules used within the PlantCare+ system, specifically between the plant pot and rotating base.

Function
Provides communication between ESP32-based modules within the PlantCare+ system using ESP-NOW protocol over 2.4 GHz.

Features

- Enables wireless communication between system modules.
- Based on the ESP32 modules used in the PlantCare+ design.
- Supports communication required for extended module functionality.
- Required for features such as manual rotation mode.

Operating Principle
The module establishes a wireless link between compatible ESP32-based subsystems, allowing control data and status information to be exchanged during operation.

Integration

- Communication method: proprietary implementation using ESP32 modules.
- System requirement: standard PlantCare+ device.
- Dependent modules listed in the table include Light Intensity Tracking and the OLED Display Suite.

Use-Case

Intended for system configurations where wireless data exchange is required between PlantCare+ modules, including communication needed for user-controlled and sensor-assisted functions

6. Parameters of Extensible Module Components

Module	Parameter	Value
DHT11 Temperature + Humidity [1]	Temperature range	0°C to 50°C
	Temperature accuracy	±2°C
	Temperature resolution	1°C
	Humidity range	20% to 80% RH
	Humidity accuracy	±5% RH
	Humidity resolution	1% RH
	Sampling rate	1 Hz
	Operating voltage	3.3V – 5.5V
	Operating current	0.3 mA (active), 60 µA (standby)

Module	Parameter	Value
SSD1306 OLED [2]	Display resolution	128 × 64 px
	Operating voltage	3.3V – 5V
	Viewing angle	>160°
	Power consumption	0.04W (normal operation)
	Operating temperature	−30°C to +80°C
ESP-NOW Communication [3]	Frequency band	2.4 GHz
	Range (open field)	Up to 220 m
	Range (indoor)	40–100 m
	Max TX power	+21 dBm

All parameters ranges listed above are valid for operation and performance within the range of operation of the standard PlantCare+ device.

The accuracy of components used also fits within accuracy range of standard PlantCare+ device.

References

[1] DHT11 Temperature and Humidity Sensor Datasheet

Available at: https://www.mouser.com/datasheet/2/758/DHT11-Technical-Data-Sheet-Translated-Version-1143054.pdf?srsltid=AfmBOoquZYLwGJGmZc0yATYoLKxFVA7eIrUJL2Osdeu-KOwp4shplKBe

[2] 128 x 64 OLED Module SSD1306 (SPI) Datasheet

Available at: https://cdn-shop.adafruit.com/datasheets/SSD1306.pdf

[3] ESP-NOW Communication Protocol

Available at: https://docs.espressif.com/projects/esp-idf/en/stable/esp32/api-reference/network/esp_now.html